WORKSHEET — UNIT II

Python Tools for Data Handling

Course: Data Science with Python **Unit:** II (10 Hrs)

Name: _____ **Roll No:** _____ **Date:** _____

Topics covered: Tkinter GUI • Jupyter • REST APIs • Web scraping • SQL & databases

Section 1 — Key Concepts (Quick Revision)

Term	Simple Meaning	Example
Tkinter	Python's built-in library for making desktop windows.	`import tkinter as tk`
Widget	A part of a GUI, such as a label, button or entry box.	`tk.Button(root, text="OK")`
Jupyter Notebook	Write code, see output and add notes in one page.	`Shift + Enter runs a cell`
API	A way for one program to talk to another program.	`weather API, maps API`
REST API	A web service you call over HTTP using GET or POST.	`GET = read, POST = send`
requests	Python library used to call APIs.	`requests.get(url)`
JSON	Text format for data, written as key : value pairs.	`{"name": "Ravi", "age": 20}`
Authentication	Proving who you are, for example with an API key.	`headers = {"Authorization": key}`
Web scraping	Collecting data from a web page using code.	`get HTML, then pick out data`
BeautifulSoup	Library that reads HTML and finds tags.	`soup.find("h1")`
SQL	Language used to work with database tables.	`SELECT * FROM students`
sqlite3 / SQLAlchemy	sqlite3: small built-in database. SQLAlchemy: use Python objects instead of raw SQL.	`sqlite3.connect("data.db")`

Section 2 — Multiple-Choice Questions

Choose the correct option and circle the letter.

1. **Which Python library is used to build simple desktop GUI windows?**
   - **A.** NumPy **B.** Pandas
   - **C.** Tkinter **D.** Requests
2. **Which tool lets you mix code, output and text notes in one document?**
   - **A.** Jupyter Notebook **B.** Tkinter
   - **C.** SQLite **D.** BeautifulSoup
3. **Which HTTP method is normally used to get (read) data from an API?**
   - **A.** POST **B.** DELETE
   - **C.** PUT **D.** GET

4. **Which Python library is used to send HTTP requests to a web API?**
   A. tkinter **B.** requests
   C. sqlite3 **D.** matplotlib

5. **Which format do most REST APIs use to send back data?**
   A. CSV **B.** JSON
   C. XLSX **D.** DOCX

6. **Which library is used to read and search HTML pages when web scraping?**
   A. SQLAlchemy **B.** Tkinter
   C. BeautifulSoup **D.** NumPy

7. **Which SQL command is used to read records from a table?**
   A. INSERT **B.** SELECT
   C. UPDATE **D.** DELETE

8. **Which built-in Python module gives you a small file-based database?**
   A. sqlite3 **B.** requests
   C. tkinter **D.** json

9. **Which SQL command adds a new row to a table?**
   A. SELECT **B.** UPDATE
   C. DELETE **D.** INSERT

10. **What do you send to prove who you are when calling a protected API?**
    A. A widget **B.** An API key
    C. A query **D.** A table

Section 3 — Fill in the Blanks

Word bank: Tkinter requests JSON BeautifulSoup sqlite3

1. _____ is Python's standard library for building simple desktop GUIs.
2. The _____ library is used to make HTTP requests to REST APIs.
3. _____ is a simple text format used to exchange data with web APIs.
4. _____ is a Python library used to read HTML pages while web scraping.
5. The _____ module gives a simple file-based database in Python.

Section 4 — True or False

1. Jupyter Notebook lets you mix code, text and output in one document. **(True / False)**
2. The POST method is used only to read data, never to send it. **(True / False)**
3. Tkinter is used for web scraping. **(True / False)**
4. BeautifulSoup is used to read and search HTML pages. **(True / False)**
5. JSON is a text format used to store and send data. **(True / False)**

Section 5 — Match the Following

Write the correct letter from Column B next to each item in Column A.

No.	Column A	Ans.	Column B
1	Tkinter		a. Reads and searches HTML pages
2	Jupyter		b. Text format for API data
3	requests		c. Builds desktop windows (GUI)
4	BeautifulSoup		d. Sends HTTP requests
5	SQLite		e. Notebook for code and notes
6	JSON		f. Small file-based database

Section 6 — Short-Answer Questions

Answer each question in 1–2 sentences.

1. **What is Jupyter Notebook used for?**

2. **Write one difference between a GET and a POST request.**

3. **What is BeautifulSoup used for in web scraping?**

4. **What is SQLite? Give one reason to use it.**

5. **Give one example of authentication when calling an API.**

Section 7 — Practical Programs

Write and run each program in Python. Fill the blanks (___) using the hint.

Practical 1: Text entry and button (Tkinter)

1. Create a Tkinter window.
2. Add an Entry box and a Button.
3. When the button is clicked, show the typed text in a Label.

Hint:

```
import tkinter as tk

def show():
    label.config(text=___)        # get the text from the entry box

root = tk.Tk()
entry = tk.Entry(root)
entry.pack()
button = tk.Button(root, text="Show", command=___)   # call the function
button.pack()
label = tk.Label(root, text="")
label.pack()
root.mainloop()
```

Expected output: *A window. After you type text and click Show, the text appears in the label.*

Practical 2: Get data from an API (requests)

1. Send a GET request to the API link given below.
2. Read the JSON response.
3. Print the "title" and "completed" fields.

Hint:

```
import requests

url = "https://jsonplaceholder.typicode.com/todos/1"
response = requests.get(___)
data = response.___()      # converts JSON to a dictionary
print(data["title"])
print(data["completed"])
```

Expected output: *The title text and True/False for completed. (Needs an internet connection.)*

Practical 3: Small SQLite database

1. Connect to a SQLite database using sqlite3.
2. Create a table "students" with name and marks columns.
3. Insert two records and print all rows.

Hint:

```
import sqlite3

conn = sqlite3.connect("school.db")
cur = conn.cursor()

cur.execute("CREATE TABLE IF NOT EXISTS students (name TEXT, marks INTEGER)")
cur.execute("INSERT INTO students VALUES ('Asha', 85)")
cur.execute("INSERT INTO students VALUES ('Ravi', 78)")
conn.commit()

cur.execute("SELECT ___ FROM students")      # * means all columns
for row in cur.fetchall():
    print(row)
conn.close()
```

Expected output: *('Asha', 85) and ('Ravi', 78) printed as rows.*

Instructor Answer Key

Section 2 — MCQ

Q	Answer	Why
1	C	Tkinter is Python's standard GUI library.
2	A	Jupyter Notebook mixes code, output and text.
3	D	GET is used to fetch (read) data.
4	B	The requests library makes HTTP calls simple.
5	B	JSON is the most common format for API responses.
6	C	BeautifulSoup reads and searches HTML.
7	B	SELECT reads records from a table.
8	A	sqlite3 comes with Python and uses a single file.
9	D	INSERT adds a new row to a table.
10	B	An API key (or token) proves who the caller is.

Section 3 — Fill in the Blanks

1. Tkinter 2. requests 3. JSON 4. BeautifulSoup 5. sqlite3

Section 4 — True or False

1. True 2. False 3. False 4. True 5. True

Section 5 — Match the Following

1 → c 2 → e 3 → d 4 → a 5 → f 6 → b

Section 6 — Short Answers (Key Points)

- It is used to write and run code, see the output, and add notes, all in one place.
- GET reads data from a server. POST sends new data to the server.
- It reads the HTML of a page so that tags and text can be found and extracted.
- SQLite is a small database stored in one file. It needs no server, so it is easy to set up.
- Sending an API key or token in the request header.

Section 7 — Practical Programs (Sample Solutions)

Practical 1

```
import tkinter as tk

def show():
    label.config(text=entry.get())

root = tk.Tk()
entry = tk.Entry(root)
entry.pack()
button = tk.Button(root, text="Show", command=show)
button.pack()
label = tk.Label(root, text="")
label.pack()
root.mainloop()
```

Practical 2

```
import requests

url = "https://jsonplaceholder.typicode.com/todos/1"
response = requests.get(url)
data = response.json()
print(data["title"])
print(data["completed"])
```

Practical 3

```
import sqlite3

conn = sqlite3.connect("school.db")
cur = conn.cursor()
cur.execute("CREATE TABLE IF NOT EXISTS students (name TEXT, marks INTEGER)")
cur.execute("INSERT INTO students VALUES ('Asha', 85)")
cur.execute("INSERT INTO students VALUES ('Ravi', 78)")
conn.commit()
cur.execute("SELECT * FROM students")
for row in cur.fetchall():
    print(row)
conn.close()
```